# Accelerating Power Flow Calculations in LV Networks Using Physics-Informed Graph Neural Networks

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***Abstract*—Precise calculations of network states and parameters are crucial for the modeling, monitoring, and operation of the distribution grid. AC power flow calculations are the preferred method for determining grid operating limits and assessing system hosting capacity. Traditional numerical power flow solvers are computationally intensive for large networks due to their iterative nature. In this context, we propose a physics-informed graph neural network technique as a fast alternative to the iterative power flow calculation methods. Our approach leverages the inherent physical knowledge of power systems through the integration of different physical constraints inside the model architecture and loss function. By incorporating physical laws such as Kirchhoff's law and active power balance as prior knowledge, the proposed method maintains high accuracy while significantly reducing computational time. The results show that the proposed model achieves better performance and higher generalizability than off-the-shelf data-driven approaches.**

***Index Terms*—Distribution grids, power flow analysis, physics-informed neural networks.**

## I. Introduction

In power flow (PF) analysis, the primary objective is to determine the system's voltages and currents under specific load conditions. The distribution system operator needs to periodically analyze these network parameters to assess operating conditions and identify any problematic parts within the network. Traditional numerical power flow (PF) solvers, such as the Newton-Raphson method, are computationally expensive for large networks, with each minor change in grid configuration requiring a new calculation.

Deep learning (DL) approaches can provide faster PF solutions by learning intricate patterns from historical data [1]. However, these methods often struggle to capture the complex dynamics and constraints of electrical networks and typically require extensive training data. Most DL-based approaches proposed for solving the PF problem utilize models based on fully connected neural networks (FCNN) [2]. While FCNNs are simpler to implement, they have limited ability to adapt to changes in network topology. In contrast, graph neural networks (GNN) leverage the inherent structure of the network, making them adaptable to grids of varying sizes and configurations. A comprehensive review in [3] explores the application of GNNs in power systems, particularly highlighting their effectiveness in fault detection and PF analysis tasks. In [4], a Graph Attention Network is employed to perform the PF analysis. This helps in improving the learning process by assigning varying levels of importance to individual buses, thereby enabling the model to prioritize critical buses.

Most DL-based methods used in power systems are largely agnostic to the underlying physical principles. This oversight can lead to imprecise estimates, particularly when working with limited data or facing conditions that differ from those observed during the training phase. This underscores the need for advanced methods that combine DL-based approaches with domain-specific knowledge to improve prediction accuracy. In this context, physics-informed neural networks (PINN) approaches present a promising solution for solving complex problems in various scientific domains including the power system. PINNs leverage system equations to guide neural network learning, ensuring that predictions comply with underlying physical laws. Additionally, PINNs require less training data due to the integration of a physics-based regularization term, which effectively confines the solution space to physically plausible outcomes.

Previous research has investigated the use of PINNs in various power system tasks, including parameter estimation, dynamic state estimation, and fault detection [4], [5]. The authors in [6] propose a PINN approach that integrates the AC PF equation into the model's loss function to perform state estimation for partially observable low-voltage (LV) networks. However, this approach requires model retraining for new topologies which emphasizes the need for more flexible and adaptable solutions. Most of the current literature on PINN focuses on transmission networks, while their effectiveness in LV network problems remains largely unexplored. Therefore, it is essential to develop faster and more generalizable PINN-based methods specifically tailored for three-phase LV networks.

In this paper, we propose a novel power flow-informed graph neural network (PFiGNN) method to achieve a faster approximation of the PF problem by effectively determining voltages and currents in the LV network. Our objective is to maximize the benefits of data-based learning while ensuring compliance with the underlying physical laws.

The main contributions of this paper are:

- We developed a physics-informed graphical learning method that combines data-based and physics-based learning to efficiently and accurately perform PF calculations for three-phase LV networks.
- The proposed approach incorporates physical constraints into both the model's architecture and its loss function.
- The model can handle grids with varying topologies while maintaining high accuracy.

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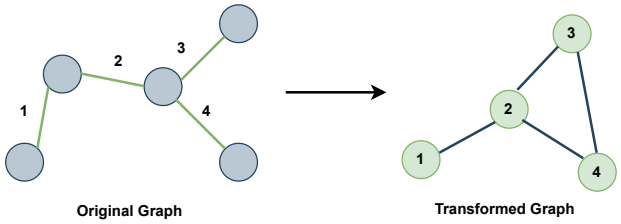



Fig. 1: Simple illustration of the transformation from bus graph to line graph representation.

The paper is structured as follows: Section II provides a description of the proposed methodology employed in this study. Experimental results are discussed in detail in Section III, followed by a conclusion in Section IV.

## II. Methodology

This section discusses how the power flow formulation is interpreted within the context of GNN algorithms and presents the proposed PFiGNN model.

### A. Interpreting Power Flow as a GNN Problem

Electrical networks are inherently graph-like in their structure which makes GNN a suitable choice for their modeling. The network topology can be represented as an undirected graph $G(N, E)$, where the set of graph nodes $N$ corresponds to network buses and the set of edges $E$ represents power lines. The node features encompass various bus attributes, including voltage magnitude and active and reactive power injections. Similarly, the edge features contain the resistance and reactance values for each line. In GNN, message passing [7] serves as a communication protocol. It allows nodes to iteratively update their features by aggregating information from neighboring nodes. Intuitively, the fundamentals of PF in LV networks naturally align with the message passing strategy employed by GNN. Given active $\boldsymbol{P}_i$ and reactive $\boldsymbol{Q}_i$ power injections linked to bus $i$ of the network, along with admittance matrix $\boldsymbol{Y}$ as the input features, our objective is to determine the output bus voltages $\boldsymbol{V}_i$ and line currents $\boldsymbol{I}_{ij}$, thereby transforming the PF problem into a node/edge regression task.

In this work, we propose an alternative dual representation of network graph. This representation leverages both the original network graph and a transformed line graph generated by a new *graph transformation layer*. Fig. 1 shows the transition from the original bus graph to its corresponding line graph. In the original graph, buses are connected by electrical lines, whereas in the transformed graph, lines become the vertices and buses form the edges. This dual graph representation, inspired by the idea presented in [8], offers a more comprehensive view of the network and enables targeted optimization for both bus connectivity and line configurations.

### B. Proposed Power Flow-informed GNN Model

The proposed model, termed PFiGNN, is based on integrating the physics of LV networks into the GNN algorithm. The architecture includes a *Graph Attention Network* (GAT) layer, a specialized power flow computation layer and a graph transformation layer. Input features $\boldsymbol{X}$ including power injections and line impedances are processed through two parallel paths as shown in Fig. 2. Initially, the GAT layer processes these input features $\boldsymbol{X}$ and produces a new set of node features (or embeddings) $\boldsymbol{X'}$ using the attention mechanism $a$. This mechanism enables each node to rank its neighboring nodes in the graph by assigning them different attention coefficients that reflect their individual relevance. The attention coefficients $\alpha_{ij}$ for node $i$ with respect to its neighboring node $j$ are given as:

$$e_{ij} = a(\mathbf{w}\boldsymbol{X}_i, \mathbf{w}\boldsymbol{X}_j), \quad \alpha_{ij} = \text{softmax}_j(e_{ij}) \tag{1}$$

where $\mathbf{w}$ represents the learnable weight matrix that transforms input feature vectors into a new feature space. The term $e_{ij}$ represents the raw attention score, which is subsequently normalized using the softmax function to yield the attention coefficients $\alpha_{ij}$. These coefficients quantify the influence of neighboring node $j$ on node $i$. The final output features of node $i$ through the GAT layer are given as:

$$\boldsymbol{X}'_i = \sigma\left(\sum_{j\in\mathcal{N}(i)} \alpha_{ij}\boldsymbol{X}_j\right) \tag{2}$$

where $\sigma$ denotes the $ReLU$ activation function, which introduces non-linearity into the model to capture complex patterns. The GAT layer enables the proposed model to focus on the most relevant aspects of the graph to produce accurate results.

The *Power Flow Computation Layer* in the model architecture computes the approximate active and reactive power

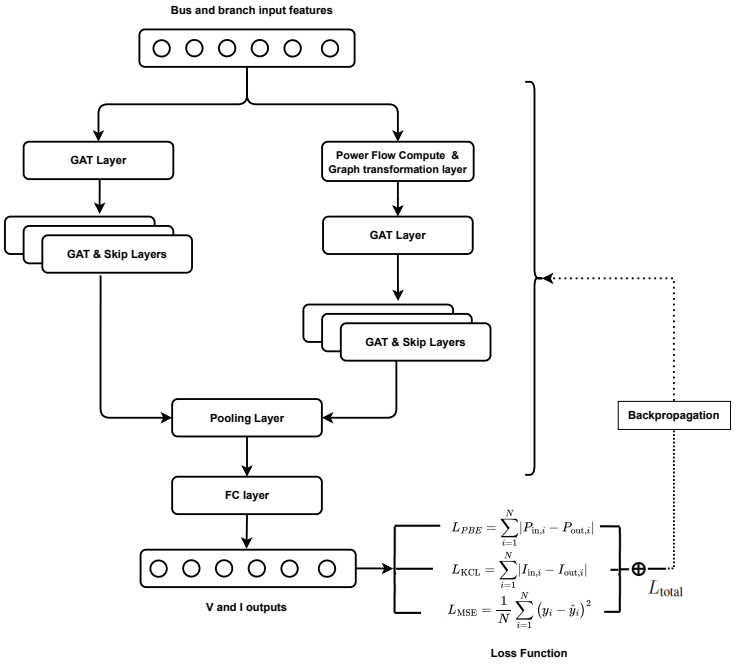



Fig. 2: Proposed PFiGNN Model

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$PQ$-flows across the edges of the grid graph. This computation leverages the mathematical formulations of nodal power balance given in [9] through the following equations:

$$\sum_{k:j\to k} P_{jk} \simeq P_{ij} + p_j \tag{3}$$

$$\sum_{k:j\to k} Q_{jk} \simeq Q_{ij} + q_j \tag{4}$$

The equations 3 and 4 state that the sum of all power flows from node $j$ to its connected nodes $k$ is approximately equal to the power flowing into node $j$ from node $i$ plus the power injected at node $j$. To compute the line $PQ$-flows based on these equations, we utilize a backward sweep methodology where we accumulate power demands from terminal nodes towards their parent nodes. The power flow computation layer systematically allocates these line $PQ$-flows to the corresponding edges of the original graph. This enables the direct incorporation of the fundamental principle of PF equilibrium into the model architecture. Next, the graph transformation layer assigns the $PQ$-flows to the vertices of the new line graph and passes them as input to the subsequent GAT layer. This allows the right branch of the model to use the line graph representation, while the left branch uses the original graph.

GAT layers primarily focus on learning node-specific features. To capture global features from the entire graph, we incorporate a mean pooling layer that aggregates information from the final GAT layers across both model branches. This allows the model to learn both local (i.e. node-level) and global (i.e. network-level) features. To mitigate the oversmoothing issue in the GNNs [8], where node representations become increasingly similar as model depth increases, we employ skip connections. These connections allow information from earlier layers to flow directly to deeper layers which helps in preserving unique features for each node. A fully connected (FC) layer is added to get the final voltage and current outputs.

### C. Physics-informed Loss Function

At the core of our PFiGNN model lies the implementation of a physics-informed loss function. We incorporated different physical constraints into the model loss function which guides it to produce physically plausible solutions. In this study, we implemented constraints based on Kirchhoff's Current Law (KCL) and nodal power balance into the model's loss function using algebraic equations as illustrated in Fig. 2. During training, the model is optimized to minimize both the discrepancy between predicted and true values, as well as the violation of the imposed physics-based constraints. This optimization is achieved through backpropagation, a technique that iteratively adjusts the model parameters to minimize the loss function. The overall loss function is expressed as:

$$L_{\text{total}} = L_{\text{MSE}} + \alpha \cdot L_{\text{KCL}} + \beta \cdot L_{\text{PBE}} \tag{5}$$

where $L_{\text{MSE}}$ represents the data-driven part of the loss function. The physical term $L_{\text{KCL}}$ quantifies the KCL constraint adherence, which validates that the sum of currents entering and leaving a node is zero. The second physical term $L_{\text{PBE}}$ represents the total active power balance error, calculated as the sum of absolute differences between incoming and outgoing active power at each node in the network. The active power is derived from the predicted voltages and currents, i.e. $P = \hat{V}\hat{I}\cos(\theta)$. To balance the contribution of physical loss terms during training, loss weights $\alpha$ and $\beta$ are introduced. This prevents any single physical loss term from dominating the overall training process. By including these physical constraints in the objective function, the model's learning process is guided toward solutions that are consistent with the characteristics of electrical networks.

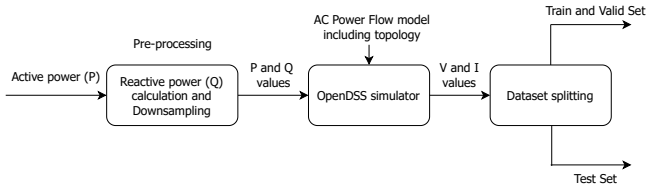



Fig. 3: Overview of the data preparation scheme

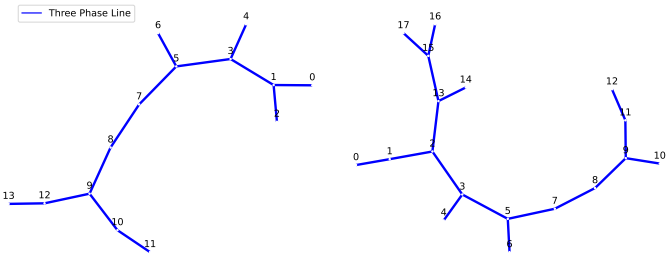
## III. Experimental Setup and Results

We evaluated the proposed model on a non-synthetic European-style LV grid documented in [10]. For our analysis, we selected three feeders of varying complexity from the grid dataset: the 14-bus (`case14`), 16-bus (`case16`) and 18-bus (`case18`) configuration. Fig. 4 illustrate the layouts of the 14-bus and 18-bus feeders. Active power load data is collected from the LV network of Flanders, Belgium over a period of 30 days and subsequently downsampled to 30-minute resolution. Reactive power profiles are generated by assuming a random power factor within the range of $0.95$ to $0.98$. Using downsampled power injection values and known network characteristics, we obtained target voltage and current values by simulating PF in OpenDSS software [11]. Fig. 3 illustrates the data preparation pipeline for model training. The dataset consisted of a total of $30\times48 = 1440$ PF snapshots. We divided it into three distinct subsets: $60\%$ for training, $20\%$ for validation and $20\%$ for testing, to facilitate the model training and evaluation process. Moreover, the dataset was standardized using per unit (p.u.) normalization using base apparent power and voltage levels. Hyper-parameter tuning is beyond the scope of this study; therefore, we used standard parameters from the literature, including the *Adam* optimizer [12] with a learning rate of $0.001$. We implemented the model using the PyTorch and PyTorch Geometric deep learning libraries.

### A. Numerical Results

We compared the performance of the proposed PFiGNN model with other DL approaches under various conditions. The models used for comparison include standard FCNN and standard GNN, both of which are physics-agnostic. Furthermore, we introduced a physics-informed variant of the FCNN model,

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(a) `case14` (b) `case18`

Fig. 4: Illustration of 14-bus and 18-bus test feeders layout.

TABLE I: Comparing Different Models based on Denormalized MSE Scores for Phase-I Voltages and Currents.

| Method | Fixed Grid Topology | | Multiple Grid Topologies | |
|---|---|---|---|---|
| | **Voltage** | **Current** | **Voltage** | **Current** |
| **Standard FCNN** | 0.151 | 1.751 | 0.961 | 4.160 |
| **Standard GNN** | 0.480 | 2.653 | 0.771 | 3.622 |
| **PIFCNN** | **0.140** | **1.675** | 0.928 | 3.934 |
| **Proposed PFiGNN** | 0.492 | 2.528 | **0.602** | **3.358** |

termed PIFCNN, by replacing the conventional loss function with a physics-informed loss function. This adjustment was made to evaluate how incorporating physical constraints in FCNN models affects its performance. We evaluated model accuracy in scenarios where the grid topology was fixed and in scenarios where the models were applied across multiple grids with different topologies. Table I displays the denormalized mean squared error (MSE) values for voltage and current predictions across both experimental scenarios.

*1) Fixed Grid Topology:* In the first experiment, we tested the proposed models on the same grid sample on which it was trained. For this task, samples of the `case18` grid are considered. This is a less challenging case in which the models only need to adapt to variation in values of a specific set of buses and lines, but not to changes in the underlying network topology. Table I shows that in a fixed topology scenario, the physics-informed FCNN model outperformed the proposed PFiGNN model, achieving approximately $50\%$ lower voltage errors and $29\%$ lower current errors. The advantage of the FCNN-based models lies in their ability to consistently learn fixed patterns from constant grid topologies, allowing them to optimize their weights more efficiently than the GNN models. Additionally, the added complexity of the proposed model increases the risk of overfitting in fixed topology scenarios. The boxplot in Fig. 6 displays the distribution of voltage values obtained by the proposed model for the `case18` grid.

*2) Generalization Across Multiple Grid Topologies:* The proposed model is capable of generalizing across grids of varying sizes and topologies. This allows a model trained on a specific grid topology to be effectively applied to new grids, regardless of their configuration or size. In the second experiment, the models are trained on samples from all three

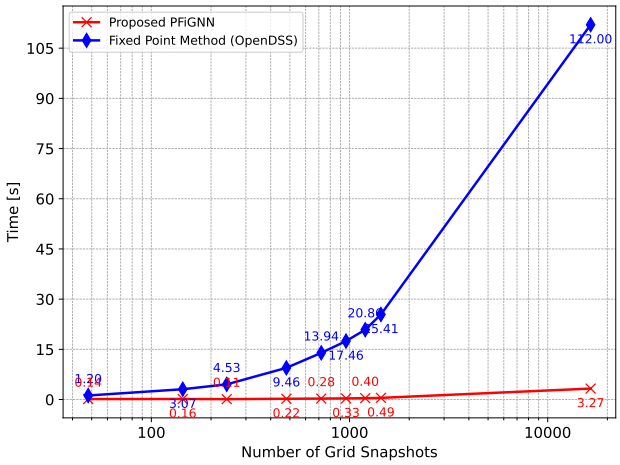



Fig. 5: Comparison of model evaluation times

feeders: `case14`, `case16`, and `case18` to evaluate their generalization performance. Unlike GNNs, the FCNN model cannot directly handle input data with varying dimensions or shapes. Therefore, to enable processing by the FCNN model, the input data is zero-padded to match the dimensions of the largest grid sample in the dataset. When evaluated on samples from three distinct feeders, the proposed PFiGNN model outperformed the FCNN model, achieving $35\%$ lower voltage and $19\%$ lower current errors. The physical loss function enables the proposed model to closely adhere to fundamental power system principles which results in more generalizable and reliable predictions across various network topologies. However, in this experiment, we observed slightly higher errors overall, particularly in current predictions. This can be attributed to the unique configurations and characteristics of the feeders, which introduce additional modeling complexity.

### *B. Time Complexity*

The main advantage of the proposed method over traditional numerical solvers is its reduced time complexity, which is crucial for performing large-scale grid simulations. We conducted a comparative evaluation of the computational times between the proposed model and the fixed point current injection method. The fixed point method, an iterative approach for solving nonlinear equations, is effective in PF analysis due to its efficiency and reliable convergence. For comparison purposes, we evaluated multiple grid snapshots with varying load demands using `case14` grid. While the fixed point method's computation time increases linearly with grid snapshots, the proposed model's evaluation time remains nearly constant, as shown in Fig. 5. This distinction highlights our model's computational efficiency by removing the need for iterative convergence processes typical to numerical solvers. Notably, the final point in the plot shows the proposed model computing $16,464$ grid snapshots in a few seconds, effectively

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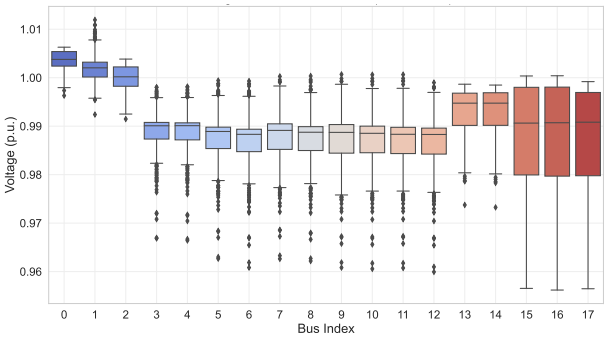



Fig. 6: Boxplot showing the bus voltage predictions for `case18` branched feeder. Voltage decreases with increasing distance from the source node due to a higher voltage drop.

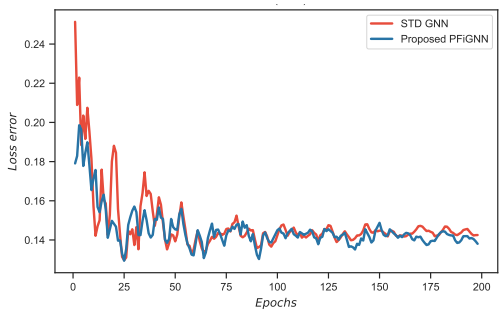
simulating nearly a full year of PF estimates. The comparison was performed on an Intel Core i7 (16 CPU cores) machine.

### C. Assessment of Model Physical Consistency

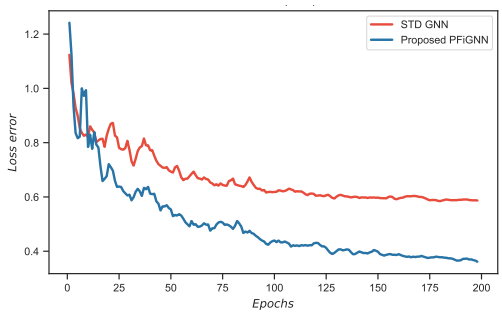
The purpose of using a PINN-based model is not only to minimize voltage and current prediction error but to also learn physical constraints as an auxiliary objective. Therefore, it is essential to evaluate the model's performance across both data-driven and physics-based learning tasks. To assess the impact of the physics-informed loss terms on the learning process, we compared the validation loss curves of the standard GNN and the proposed PFiGNN models, as shown in Fig. 7. In the physics-agnostic standard GNN model, network training relies exclusively on the data-driven MSE loss term, with the additional physical loss term (i.e. KCL error) calculated solely for monitoring purposes. Conversely, the PFiGNN model incorporates both the MSE and KCL loss terms during the training process. As illustrated in Fig. 7a, the validation MSE loss of the proposed PFiGNN model shows a slight improvement over the standard GNN model. However, the proposed model significantly outperforms the physics-agnostic GNN model in learning the KCL constraint, as shown in Fig. 7b. Consequently, in out-of-sample test scenarios, the standard GNN model is more likely to violate fundamental physical principles, such as KCL, leading to physically inconsistent predictions and reduced generalization performance.

## IV. Conclusion

We presented a novel PINN-based approach for faster power flow calculations in the LV network by embedding fundamental power system principles into the GNN model. The incorporation of physical constraints acts as a regularization mechanism, reducing the risk of overfitting and improving the model's ability to generalize across a range of network topologies. Results showed that our model outperformed benchmark DL-based approaches in both accuracy and convergence. It also demonstrates superior computational efficiency compared to traditional numerical approaches. In future work, we plan to evaluate the model in limited observability scenarios and perform hyperparameter tuning to improve its performance.




(a) MSE Error




(b) KCL Error

Fig. 7: Comparison between the loss components of physics-aware and physics-blind models.

## References

[1] V. Bassi, L. F. Ochoa, T. Alpcan, and C. Leckie, "Electrical model-free voltage calculations using neural networks and smart meter data," *IEEE Transactions on Smart Grid*, 2022.

[2] A. S. Zamzam and K. Baker, "Learning optimal solutions for extremely fast ac optimal power flow," in *2020 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*, pp. 1–6, IEEE, 2020.

[3] W. Liao, B. Bak-Jensen, J. R. Pillai, Y. Wang, and Y. Wang, "A review of graph neural networks and their applications in power systems," *Journal of Modern Power Systems and Clean Energy*, vol. 10, pp. 345–360, 2021.

[4] A. B. Jeddi and A. Shafieezadeh, "A physics-informed graph attention-based approach for power flow analysis," in *2021 20th IEEE International conference on machine learning and applications (ICMLA)*, pp. 1634–1640, IEEE, 2021.

[5] L. Pagnier and M. Chertkov, "Physics-informed graphical neural network for parameter & state estimations in power systems," *arXiv preprint arXiv:2102.06349*, 2021.

[6] J. Ostrometzky, K. Berestizshevsky, A. Bernstein, and G. Zussman, "Physics-informed deep neural network method for limited observability state estimation," *arXiv preprint arXiv:1910.06401*, 2019.

[7] J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals, and G. E. Dahl, "Neural message passing for quantum chemistry," in *International conference on machine learning*, pp. 1263–1272, PMLR, 2017.

[8] J. B. Hansen, S. N. Anfinsen, and F. M. Bianchi, "Power flow balancing with decentralized graph neural networks," *IEEE Transactions on Power Systems*, 2022.

[9] M. Baran and F. Wu, "Network reconfiguration in distribution systems for loss reduction and load balancing," *IEEE Transactions on Power Delivery*, vol. 4, 1989.

[10] A. Koirala, P. Arboleya, B. Mohamed, and L. Ramon, "Non-Synthetic European Low Voltage Test System," *International Journal of Electrical Power Energy Systems*, vol. 118, 12 2019.

[11] EPRI, "OpenDSS – Open Distribution System Simulator." [Online]. Available: http://smartgrid.epri.com/SimulationTool.aspx.

[12] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *CoRR*, vol. 1412.6980, 2014.

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